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Next-Generation Sequencing–Based Molecular Profiling of Conjunctival Squamous Cell Carcinoma and Its Potential Application for Therapy

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Objective: Targeted next-generation sequencing–based genomic and transcriptomic analyses of conjunctival squamous cell carcinoma (cSCC) samples using a panel of >1700 cancer-related genes.

Design: Prospective case series.

Participants: Twenty patients with invasive cSCC consecutively managed at an academic ocular oncology setting.

Methods: Integrative exome and transcriptome analysis of fresh-frozen tumor and matching normal samples.

Main Outcome Measures: Molecular characterization of invasive cSCCs (for somatic mutations, tumor mutation burden (TMB) and signatures, structural variations, viral transcripts, outlier gene expression) and its potential clinical applications.

Results: Of the 20 invasive cSCCs, only 2 were positive for human papillomavirus (HPV). All 18 HPV-negative tumors had genetic alterations in *TP53* and 16 also had alterations in *CDKN2A*. The 2 HPV-positive tumors showed no alterations in these cell cycle regulators but harbored mutations in *PIK3CA*. Other frequently altered genes included *KMT2C/D* (70%), *FAT1/3* (65%), and *NOTCH1/2/3* (60%), which were implicated in both the HPV-negative and positive tumors. The average TMB was 58.41 mutations per megabase (Mut/Mb). The TMB was >20 Mut/Mb in 13 cases (65%, range: 49.3–160.8 Mut/Mb), of which 11 had ultraviolet (UV) mutational signature, 3 had APOBEC signature, and 2 had microsatellite instability. Most UV-driven tumors (8 of 11) also harbored *TERT* promoter mutations. The most recurrent large-scale copy number alterations were the deletions affecting chromosomes 3p, 9p, and 14q. Uncommon but potentially drug-targetable copy number gains were also detected affecting several oncogenes. The most frequent gene expression outlier was the aberrantly expressed *TP63* found in 19 tumors.

Conclusions: In our invasive cSCC cohort, HPV infection was not a major contributor to tumor etiopathogenesis. Our results confirmed the central role of *TP53* genetic alterations and the common presence of UV signature in the HPV-negative cSCCs. Irrespective of HPV status, other commonly observed genetic alterations included those affecting chromatin modifiers followed by Hippo or Notch pathway–related genes. Ultraviolet-driven *TERT* promoter mutations co-occurred with other driver gene alterations. The commonly observed high TMB makes immunotherapy a good treatment choice for invasive cSCC. Moreover, several cSCC-associated molecular alterations represent potentially actionable targets, while further studies are necessary to understand their roles in cSCC development and invasion.

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Ocular surface squamous neoplasia (OSSN) is the most common nonpigmented tumor affecting the conjunctiva, with an age-standardized rate of 0.26 cases per 100 000 per year.[1,2] The major risk factor for OSSN is exposure to ultraviolet (UV) B radiation from sunlight. The other risk factors include older age, male sex, immunosuppression, cigarette smoking, chemical exposure, and injury.[3–5] Human papillomavirus (HPV) high-risk serotypes such as 16, 18, and 33 are also believed to play an etiologic role in OSSN, although their prevalence seems to vary geographically.[6,7]

Patients with conjunctival squamous cell carcinoma (cSCC) can present with either noninvasive lesions (i.e., conjunctival intraepithelial neoplasia or carcinoma in situ) or invasive tumors (Fig 1). While distant metastases are not common, cSCC can be highly recurrent and cause local complications. Molecular understanding of cSCC development and progression may therefore help to develop or implement more effective and tailored treatments. Due to the rare occurrence of these tumors, however, there are only a few studies in the literature concerning comprehensive genetic analyses of OSSN/cSCC.[8–12]

Apart from 2 initial whole-exome sequencing (WES) studies that analyzed only a small number of noninvasive

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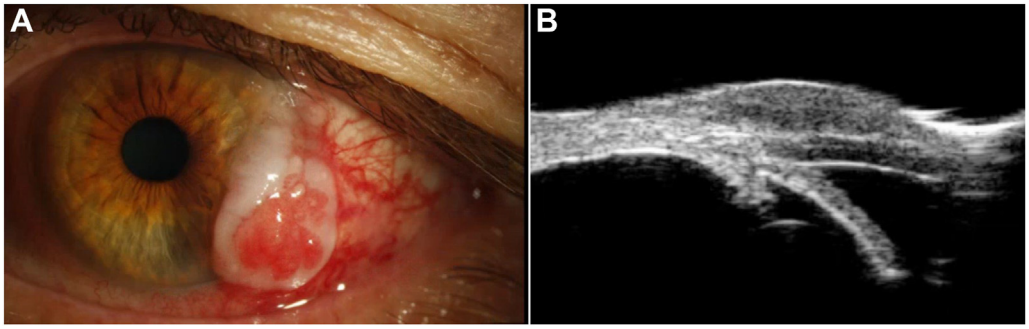


Figure 1. Invasive conjunctival squamous cell carcinoma. **A,** Slit lamp examination of the left eye showed a nodular amelanotic lesion on the temporal bulbar conjunctiva, which was histopathologically confirmed to be invasive squamous cell carcinoma. **B,** Ultrasound biomicroscopy showed the conjunctival squamous cell carcinoma had invaded the underlying sclera along the limbus.

(conjunctival intraepithelial neoplasia or carcinoma in situ) cases,[8,9] 3 studies were published examining both noninvasive and invasive archived (formalin-fixed paraffin-embedded [FFPE]) samples using either targeted sequencing (for <525 cancer-related genes)[10,11] or WES.[12] While the tumor mutation profiles were analyzed in all these 3 studies,[10–12] mutational signatures were analyzed in only 2,[10,12] tumor mutation burden (TMB) in only 1,[12] and copy number alterations (CNAs) also in only 1.[10] We previously reported the molecular features of conjunctival melanoma based on integrative exome and transcriptome analysis of a panel of >1700 cancer-related genes[13] by parallel sequencing of tumor and normal samples collected from the same patients.[14] Herein, we used this large targeted sequencing panel to similarly evaluate the molecular features of invasive cSCC and its potential applications for treatment. For this purpose, we comprehensively analyzed the sequencing data from flash-frozen samples obtained from a North American cohort for somatic mutations, mutation signatures, TMB, CNAs, and outlier gene expression. We then compared our findings with those from previous studies that evaluated invasive cSCCs obtained from similar cohorts.

Here we present our comprehensive analysis results from fresh-frozen invasive cSCC and matched normal samples with potential implications for therapy, including confirmation of several recent findings (e.g., the common observance of high TMB and UV signature, the central role of altered cell cycle regulators such as *tumor protein p53 [TP53]/cyclin-dependent kinase inhibitor 2A [CDKN2A]*, and the major role of altered chromatin modifiers such as lysine methyltransferase 2 [KMT2] genes) as well as novel findings (e.g., the common co-occurrence of UV-driven *telomerase reverse transcriptase [TERT]* promoter mutations with other driver mutations, and the recurring large deletions on chromosomes 3p, 9p, and 14q).

Methods

Prospective integrative exome and transcriptome analysis was conducted in the Translational Pathology Department, University of Michigan, using Mi-OncoSeq as described previously.[13] Mi-OncoSeq is a **C**linical **L**aboratory **I**mprovement **A**mendments (CLIA)-certified test developed by **Mi**chigan **Onco**logy **Seq**uencing Center for the next-generation sequencing (NGS)–based evaluation of >1700 cancer-related genes, by parallel analyses of tumor and normal samples collected from the same subjects.[13] Flash-frozen tumor and matched normal (cheek swab, sputum, or blood) samples collected during the surgical procedure were used for this analysis. The areas with the highest tumor content were identified by examination of hematoxylin–eosin–stained tissue sections and then processed for sequencing. Integrative exome and transcriptome analysis included the evaluation of tumors for somatic point mutations, small insertions or deletions, CNAs, viral transcripts, and outlier gene expression as described previously.[13] Microsatellite instability (MSI) was assessed by using the MSIsensor.[15] APOBEC mutational signatures were detected as described previously.[16] Ultraviolet signature was identified by the hypermutation phenotype associated with CC > TT mutations.

The study was approved by the University of Michigan Institutional Review Board and adhered to the tenets of the Declaration of Helsinki.[17] All participants provided written informed consent prior to their enrollment in the study. As part of the Mi-Oncoseq clinical sequencing program, 24 patients with cSCC were enrolled from November 2015 to March 2018. Following postsequencing quality control and exclusion of cases with low tumor purity (<20%), quality control–passed tumor and normal tissue DNA and RNA sequencing data were available for integrative analysis from a total of 20 patients with cSCC.

Results

The demographic and clinical features of the North American study cohort, comprising a total of 20 cases with quality control–passed sequencing data, are shown in Table 1. The anatomical locations of the studied tumors are depicted in Figure 2. Of these 20 cSCCs, only 2 (10%) were found to be positive for HPV according to RNA sequencing–based viral transcript analysis, which revealed high-risk HPV serotype 16 positivity in both cases.

Somatic genetic alterations were enriched in the following biological pathways and genes: the cell cycle pathway (*TP53, CDKN2A, retinoblastoma 1 [RB1], cyclin D1 [CCND1], cyclin-dependent kinase 6 [CDK6]*) in 90% of tumors, epigenetic regulation via chromatin modifiers (*lysine methyltransferase 2C/D [KMT2C/D], AT-rich interaction domain 1/2 [ARID1/2], tet methylcytosine dioxygenase 2 [TET2], DNA methyltransferase 3 alpha [DNMT3A]*) in 75% of tumors, the Hippo pathway (*FAT atypical cadherin 1/3 [FAT1/3], yes-associated protein 1 [YAP1]*) in 70% of tumors, the

Table 1. Demographics and Clinical Features of Patients with Invasive Conjunctival Squamous Cell Carcinoma

Feature	Number (%)
Age (years)	
Mean	68
Median	72
Range	44–89
Sex	
Male	16 (80%)
Female	4 (20%)
Race	
European American	19 (95%)
African American	1 (5%)
Involved eye	
Right	8 (40%)
Left	12 (60%)
Size of conjunctival involvement (mm)	
Mean	7
Median	4
Minimum–Maximum	1–25
Histopathological subtypes	
Well-differentiated SCC	12 (60%)
Moderately differentiated SCC	4 (20%)
Poorly differentiated SCC	3 (15%)
Spindle cell SCC	1 (5%)
Management outcome	
Recurrence	4 (20%)
No recurrence	16 (80%)
Systemic outcome	
Metastasis	2 (10%)
No metastasis	18 (90%)

SCC = squamous cell carcinoma.

Notch pathway (*notch receptor 1/2/3 [NOTCH1/2/3]*) in 60% of tumors, the phosphatidylinositol 3-kinase (PI3K) pathway (*PI3K catalytic subunit alpha/beta/gamma [PIK3CA/B/G], PI3K catalytic subunit type 2 alpha [PIK3C2A], PI3K regulatory subunit 1/2 [PIK3R1/2]*) in 55% of tumors, and the Ras/mitogen-activated protein kinase (RAS/MAPK) pathway (*HRas proto-oncogene, GTPase [HRAS], neurofibromin 1 [NF1], Ras p21 protein activator 1 [RASA1], Cbl proto-oncogene [CBL], fibroblast growth factor receptor 3 [FGFR3], erb-b2 receptor tyrosine kinase 2/3 [ERBB2/3]*) in 40% of tumors (Fig 3). Alterations were also detected in genes with other or multiple functions (*kelch like ECH associated protein 1 [KEAP1], GNAS complex locus [GNAS], F-box and WD repeat domain containing 7 [FBXW7], CREB binding protein [CREBBP], E1A binding protein p300 [EP300]*), of which *CREBBP* and *EP300* (each altered in 25% of tumors) are also known to act as chromatin modifiers (Fig 3).

The most commonly altered individual genes or gene groups were *TP53* in 18 of 20 tumors (90%), followed by *CDKN2A* in 16 tumors (80%), *KMT2C/D* in 14 tumors (70%), *FAT1/3* in 13 tumors (65%), *NOTCH1/2/3* in 12 tumors (60%), and PIK3C/R genes in 11 tumors (55%) (Fig 3). All of the 18 HPV-negative cases (90%) showed somatic alterations in *TP53* (Fig 3). In the 2 HPV-positive cases (10%), no somatic alterations were observed in *TP53* or other cell cycle-related genes that were found altered in the HPV-negative cases. In these 2 HPV-positive cSCCs with no somatic *TP53* alterations, the most commonly mutated genes were *PIK3CA/B* and *KMT2C/D* (Fig 3).

Promoter mutations of *TERT*, a gene playing a major role in telomere maintenance, were found in 40% of tumors (Fig 3). While most of these were hotspot mutations, interestingly, 2 tumors shared the same nonhotspot *TERT* promoter mutation, c.-138/-139CC>TT. In our cohort, all *TERT*-mutated tumors were HPV-negative, showed UV mutational signature, and harbored concurrent alterations in other driver genes.

The average TMB in our cohort was 58.41 mutations per megabase (Mut/Mb) (range 1.5–160.8 Mut/Mb). Tumor mutation burden was >20 Mut/Mb in 13 cases (65%, range: 49.3–160.8 Mut/Mb) and <20 Mut/Mb in 7 cases (35%, range: 1.5–8.3 Mut/Mb). Of the 13 cases with >20 Mut/Mb, the majority (12 of 13, 92%) showed UV mutational signature (n = 10) or MSI (n = 1), or both (n = 1) (Fig 3). All of the UV signature–positive cases were in the HPV-negative group (11 of 18, 61%). The HPV-negative tumors that did not demonstrate UV-induced damage (n = 7) were located at sites not exposed to the sun, such as under the eyelids and along the inferior fornix (Fig 2). Of the 2 cases with MSI, one had biallelic inactivation of *mutL homolog 1 (MLH1)* and the other had biallelic inactivation of *mutS homolog 2 (MSH2)*, suggesting mismatch repair deficiency. All alterations were of somatic nature. APOBEC mutational signature was detected in 3 cases

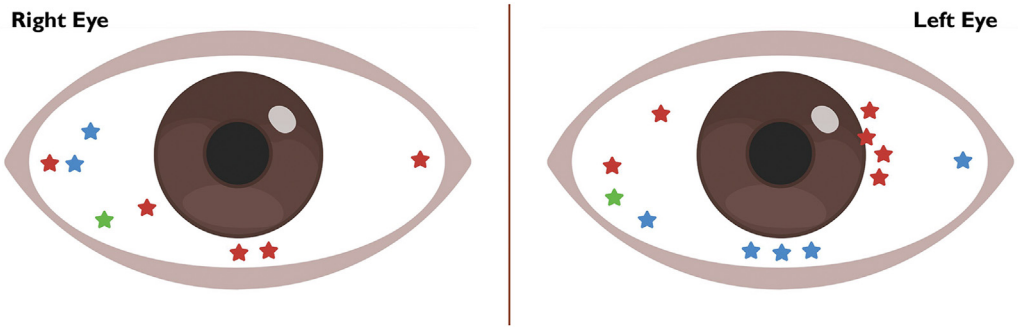


Figure 2. The locations of invasive squamous cell carcinomas over the conjunctiva. Red stars: HPV-negative tumors with UV signature; blue stars: HPV-negative tumors without UV signature; green stars: HPV-positive tumors without UV signature (created with BioRender.com). HPV = human papillomavirus; UV = ultraviolet.

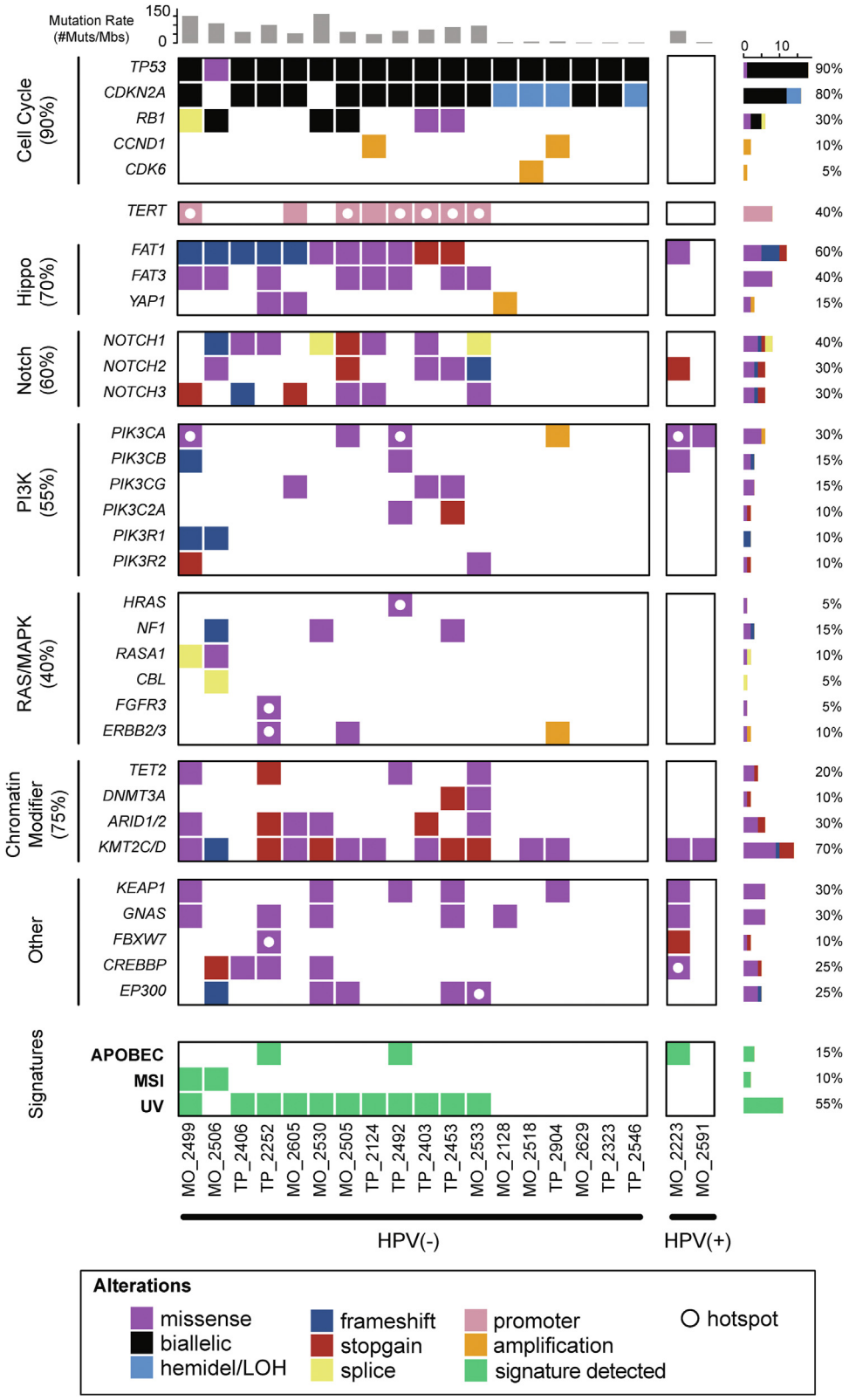


Figure 3. Oncoprint for somatic genetic alterations in our cohort of invasive conjunctival squamous cell carcinomas, grouped by HPV status. When there were several mutations in a gene, for the sake of clarity, only 1 type of mutation was selected to depict, prioritized in the following order: stop-gain, frameshift, splice, and missense. A hotspot mutation was defined as one detected in >10 patients in the Catalogue of Somatic Mutations in Cancer (COSMIC). HPV = human papillomavirus; MSI = microsatellite instability; Mut/Mb = mutations per megabase; UV = ultraviolet.

(15%), of which 2 were in the HPV-negative and 1 was in the HPV-positive group (Fig 3).

The most recurrent large-scale copy number events observed in our cohort were chromosome arm−level deletions at 3p (55%), 9p (45%), and 14q (30%) (Fig 4A). Some tumors harbored uncommon but interesting copy number gains or amplifications of potentially drug-targetable oncogenes such as *CCND1*, *CDK6*, *YAP1*, *PIK3CA*, and *ERBB2* (Figs 3 and 4B).

The most frequently detected outliers in our gene expression analysis included *tumor protein p63 (TP63)* in 19 tumors (95%), keratin (KRT) genes (*KRT1/2/5/7/10/12/13−17/19*) in 18 tumors (90%), *FAT atypical cadherin 2 (FAT2)* in 6 tumors (30%), and *cadherin 3 (CDH3)* in 5 tumors (25%).

Discussion

Our NGS-based comprehensive analysis of a North American cohort of invasive cSCCs (n = 20) has shown a common presence of high TMB and UV mutational signature in these tumors, as well as a common occurrence of genetic alterations affecting cell cycle regulators, chromatin modifiers, and Hippo or Notch pathway−related genes (Fig 3). Our cohort predominantly comprised HPV-negative cases (18 of 20, 90%) as defined by viral transcript analysis. The 2 HPV-positive tumors in our series (both positive for HPV-16) were located in the forniceal conjunctiva and lacked UV mutational signature (Figs 2 and 3). These HPV-positive cases were characterized by a common occurrence of mutations in PI3K pathway−related genes and chromatin modifiers (Fig 3). While the percentage of the HPV-positive cases was relatively lower (10%) in our cohort compared with those recently reported (>20%) in similar-sized North American or European invasive cSCC cohorts,[11,12] it was within the HPV prevalence range (7%−32%) reported for invasive cSCC according to a recent meta-analysis of 39 studies of intraepithelial or invasive cases.[7]

Based on our somatic genetic analyses, the most commonly altered genes in our cohort were cell cycle-related tumor suppressor genes *TP53* (90%) and

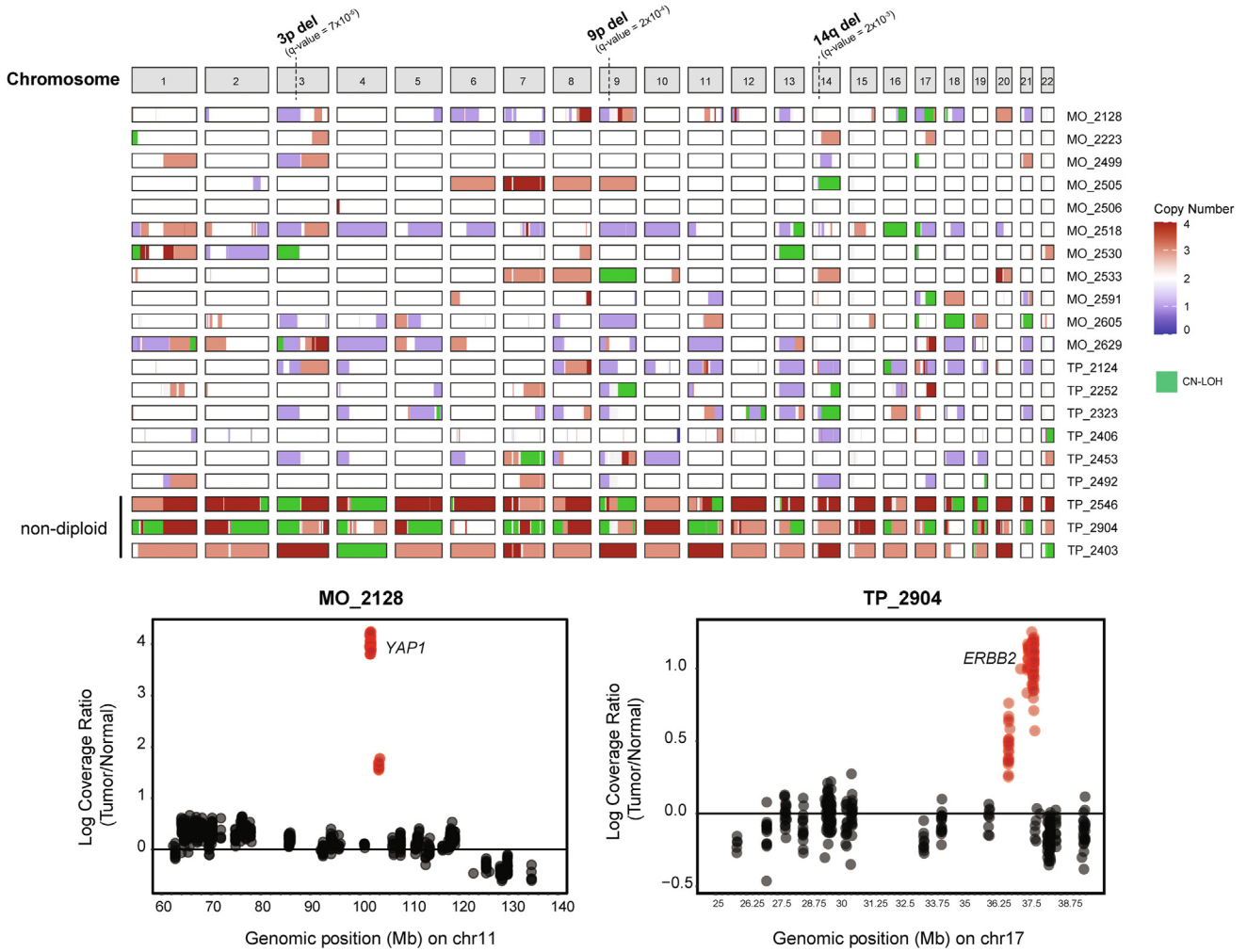


Figure 4. Structural genetic variations in our cohort of invasive conjunctival squamous cell carcinomas. Upper panel: copy number alteration heatmap for gain, loss, and copy neutral loss of heterozygosity (CN-LOH). Lower panel: examples of focal amplification that lead to overexpression.

CDKN2A (80%), which predominantly showed biallelic alterations (Fig 3). *TP53* (located on chr 17p13) encodes p53, a 53-kDa phosphoprotein that plays major roles in cell cycle control, DNA repair, and induction of senescence and cell death in response to various stress signals.[18,19] It can be regulated by *CDKN2A* (via *MDM2*) and can regulate genes such as *CDKN1A* and *NOTCH*.[19–21] *TP53* was initially implicated in OSSN/cSCC via targeted analysis of the gene or related mutation(s).[22–24] While an initial small-sized (n = 7) WES study of noninvasive cases did not report any *TP53* mutations,[8] subsequent NGS studies (using WES or targeted analysis of cancer-related genes in noninvasive or mixed cohorts) have established altered *TP53* as a major contributor to cSCC pathogenesis.[9–12] Two of these studies, which analyzed both noninvasive and invasive archived (FFPE) samples for somatic mutations, have reported *TP53* mutations in 59% or 67% of their North American (29 total, 18 invasive) or European (33 total, 25 invasive) cases.[11,12] A previous targeted NGS study from our Michigan center, which analyzed 17 of 21 invasive archived (FFPE) samples for both somatic mutations and copy number changes, has reported *TP53* alterations in 94% of North or South American cases.[10] Likewise, the *TP53* genetic alteration rate was 90% in 20 fresh-frozen invasive samples examined in the current study using both mutation and copy number analysis. Consistent with the previous reports of the enrichment of *TP53* mutations mainly in the HPV-negative cases,[11,12] all of the *TP53* alterations identified in the current study were in the HPV-negative tumors. While none of the 2 HPV-positive tumors showed genetic alterations in *TP53* in our series, HPV-related oncoproteins are known to downregulate tumor suppressor proteins p53 and **R**etino**B**lastoma **p**rotein (pRB) via direct interactions and mechanisms.[25,26]

CDKN2A (located on chr 9p21) can control the cell cycle and promote the activities of p53 and pRB by encoding the tumor suppressor proteins $p14^{ARF}$ and $p16^{INK4A}$.[27] Our study identified *CDKN2A* as the second most frequently altered gene in invasive cSCC based on both mutation and copy number analysis, at a rate (80%) much higher than that (<25%) reported in prior North American and European cohorts analyzed only for mutations.[11,12] In the previous study from our center,[10] *CDKN2A* copy number loss was frequently found in both skin and ocular SCCs, as well as in their precursor lesions. In archived invasive samples included in that study,[10] the *CDKN2A* alteration rate (86%) was very similar to that (80%) observed in fresh-frozen samples analyzed in this study, suggesting that both mutations and copy number changes of *CDKN2A* can be successfully detected in archived (FFPE) samples and copy number loss of *CDKN2A* is a major contributor to the loss-of-function of *CDKN2A* in these tumors. Interestingly, *RB1* (located on chr 13q14) genetic alterations had a relatively higher rate (30%) in our series compared with that (<20%) reported in all previous studies,[9–12] and unlike most previous reports, they were not mutually exclusive with *CDKN2A* alterations (Fig 3).

Following *TP53* and *CDKN2A*, the most commonly altered genes in our invasive cSCC series were *KMT2C/D* (70%), *FAT1/3* (65%), and *NOTCH1/2/3* (60%). Of the 2 previous studies that analyzed all these gene groups in >10 archived samples, one[12] similarly found all 3 groups among the top mutated genes, whereas the other[11] detected only the KMT2 genes among the top mutated ones. The previous study from our center[10] found only the NOTCH genes among the top mutated ones; however, the KMT2 and FAT genes were not included in that study panel which only targeted 134 cancer-related genes. While the NOTCH signaling pathway has a well-known role in carcinogenesis,[28,29] the FAT genes appear to be involved in multiple biological pathways including the Hippo and Wnt/β-catenin signaling pathways.[30,31] NOTCH and FAT gene alterations are believed to occur as early events during squamous cell carcinogenesis across various body sites.[19] In contrast, alterations of KMT2 genes and other chromatin modifiers (e.g., *CREBBP* and *EP300*) are proposed to be late events, as they have been shown to contribute to poor outcomes in skin SCC[32–35] and occur mainly in invasive cSCC according to a recent mixed cohort study.[11] This would also explain the higher rate of genetic alterations detected in these chromatin modifiers (75%) in our cohort, which comprised only invasive cases (Fig 3). Accordingly, epigenetic dysregulation seems to play an important role in the progression to invasive cSCC, and the reversible nature of epigenetic changes and recently emerging epidrugs[36] make the implicated chromatin modifiers attractive targets for the future management of advanced cSCC.

While the PIK3C/R genes were found altered in 55% of all invasive cSCCs in our series regardless of their HPV status, the *PIK3CA/B* mutations (along with the *KMT2C/D* mutations) were consistently present in all the HPV-positive tumors (n = 2), which lacked alterations in *TP53/CDKN2A* (Fig 3). Our findings are in line with those reported in a previous study,[11] which similarly detected the *PIK3CA* mutations in both the HPV-negative and positive archived samples while their rate was higher in the HPV-positive group. *PIK3CA* mutations were also detected in both noninvasive and invasive cSCCs included in 2 previous studies.[10,11] Overall, the findings from both current and previous studies regarding the mutation profiles of HPV-positive cSCCs are consistent with what is currently known about the somatic alterations occurring during HPV-induced carcinogenesis.[26] The common dysregulation of the PI3K/AKT/mTOR pathway in HPV-positive SCCs makes this pathway a promising therapeutic target in this subgroup of cSCCs.[37]

Ramberg et al[11] reported that while the PI3K pathway was most commonly affected in the HPV-positive cSCCs, the RTK/RAS pathway was more commonly affected than the PI3K pathway in the HPV-negative cases. In the previous study from our center,[10] alterations in RAS/MAPK pathway–related genes were detected in both noninvasive and invasive cSCCs, and in our current study, they were mainly found in the HPV-negative tumors (Fig 3), making this pathway a potential target for the treatment of these cases.

Previous targeted analyses of *TERT* promoter mutations have implicated them in about 30% to 50% of cSCCs, including both noninvasive and invasive cases.[38–41] Our

results are consistent with those in previous reports as we similarly found *TERT* promoter mutations in 40% of invasive cSCCs in this study. *TERT* (located on chr 5p15) is a well-known contributor to telomere maintenance and immortality in many cancers.[42] The present study, which to our knowledge was the first to analyze the *TERT* mutations concurrently with other gene alterations in cSCC, has revealed that all *TERT*-mutated tumors harbored *TP53*, *CDKN2A*, and *FAT1/3* alterations, and all but one also harbored *KMT2C/D* and *NOTCH1/2/3* alterations (Fig 3). In our series, all identified *TERT* promoter mutations were in the HPV-negative tumors that consistently showed UV mutational signature, confirming the prior suggestions based on the type of nucleotide changes detected in previous noncomprehensive cSCC studies.[38–41] *TERT* promoter mutations appear to be early events and show a similar involvement in both cSCCs and conjunctival melanomas,[43] making TERT a future therapeutic target in these conjunctival tumors given the emerging TERT vaccine and inhibitors.[42,44–46] Besides the activating promoter mutations, focal amplifications can also lead to increased *TERT* expression and activity; however, the latter was not detected in our cSCC study. Another telomere maintenance mechanism used in cancer involves the induction of alternative lengthening of telomeres pathway[43]; however, no apparent pathologic variants were identified in alternative lengthening of telomeres pathway–related genes (*ATRX* and *DAXX*) included in the panel used for our cSCC study.

In our invasive cSCC cohort, the TMB ranged from 1.5 to 160.8 Mut/Mb (average: 58.41 Mut/Mb) and was >20 Mut/Mb (range: 49.3–160.8 Mut/Mb) in 65% of cases. A recent study by Gleber-Netto et al[12] similarly found a high TMB in archived samples from their North American cohort, and even a higher TMB in their African cohort, and reported the UV-induced DNA damage and related signature as the main driver for cSCC. In comparison, the average TMB detected in our cohort was relatively lower, likely because we had fresh-frozen tumor and matching normal sample available for analysis in all our cases (which helped us exclude the germline variants) whereas the normal (germline) DNAs were available in only a subset of cases analyzed in the recent study.[12] While the high TMB was primarily driven by UV signature also in our cohort, especially in HPV-negative cases, other signatures (MSI or APOBEC) were detected in a few tumors that also had high mutation rate but no UV signature (Fig 3). The common observance of high TMB in cSCCs makes immunotherapy an attractive treatment option for these tumors.[47–50] While the UV signature was absent in HPV-positive cases in our series, the APOBEC signature appeared to be more common in these tumors compared to their HPV-negative counterparts (1 of 2 vs. 2 of 18). This observation is consistent with the previous report of enrichment of APOBEC-related signatures in the North American HPV-positive cSCC cases,[12] as well as with the overall implication of this signature in HPV-driven cancers.[26]

An initial study of CNAs in cSCC analyzed 12 samples using array-based comparative genomic hybridization and reported a common occurrence of 8p11 amplification in 75% of their samples.[51] The follow-up evaluation of this CNA in a larger cSCC cohort (using fluorescent in situ hybridization), however, suggested its presence in only <25% of the studied noninvasive and invasive cases.[52] Consistent with the latter study, we did not observe a common occurrence of 8p gains in our series but rather found recurrent 3p, 9p, and 14q losses as the most commonly occurring large CNAs in invasive cSCCs (Fig 4). Recurrent losses at these chromosomal locations were among the additional alterations detected in the initial array-based study.[51] Furthermore, we detected some uncommon but potentially drug-targetable oncogene gains and amplifications in our series such as those affecting *CCND1*, *CDK6*, *YAP1*, *PIK3CA*, and *ERBB2* (Fig 3). Similar gains and amplifications were also observed in archived (FFPE) samples analyzed in the previous study from our center.[10] Finally, our gene expression analysis has revealed *TP63* and keratin genes as the most frequently detected outliers in invasive cSCCs (in 95% and 90% of tumors, respectively). As a stem cell marker and master regulator of epidermal differentiation, p63 (encoded by *TP63*) is believed to play an important role in squamous cancer pathogenesis.[53–55]

Our study has several strengths and unique features including the evaluation of fresh-frozen invasive cSCC samples along with matching normal samples, the use of a large cancer gene panel (>1700 genes), and concurrent analyses for somatic mutations (including *TERT* promoter mutations), TMB and mutation signatures, copy number changes, viral transcripts, and outlier gene expression. The major limitation is the small size of our North American fresh-frozen invasive cSCC cohort (n = 20) due to the rarity of this cancer, although our sample size was comparable to that of American or European cohorts of archived invasive cSCCs (n = 18–25) analyzed in previous comprehensive studies.[10–12] Other limitations of our study include no assessment of the entire genome or epigenetic mechanisms (i.e., chromatin conformation, histone modifications, DNA methylation, noncoding RNA expression, and RNA modifications). While we employed RNA sequencing–based viral transcript analysis for HPV detection (expected to capture active transcription of viral genes, thus to be superior to DNA-based analysis), we recognize that confirmation of our findings by another method would also be relevant. Nevertheless, consistent with the prior literature, both HPV-positive and HPV-negative cases detected in our cohort showed the expected mutation profiles in corresponding tumors, thus providing further support to our findings.

In conclusion, in this first NGS-based comprehensive study of fresh-frozen invasive cSCCs and matched normal samples, we confirmed several findings reported in previous archived tumor studies as well as noted some novel findings. In our invasive cSCC cohort, HPV infection was not a major contributor to disease etiopathogenesis. Irrespective of the HPV status, the commonly observed genetic features included a high TMB (driven mainly by UV and less frequently by MSI or APOBEC signature) and alterations in chromatin modifiers followed by those in Hippo or Notch pathway–related genes. Group-specific "major drivers" were the genetically altered cell cycle regulators in the HPV-negative tumors and the mutated PI3K pathway–related genes in the HPV-positive tumors.

Our analyses have also revealed a common presence of UV-driven *TERT* promoter mutations co-occurring with other driver gene alterations, as well as some recurring large deletions and less commonly occurring but potentially drug-targetable oncogene gains and amplifications. While several of these cSCC-associated genetic alterations have therapeutic implications, further studies are necessary to understand their roles in cSCC-related cell proliferation, migration, and invasion. Given that both the HPV infection prevalence and cSCC genetic profiles can vary across populations and geographic locations,[7,12] a precision medicine approach is needed for better management of this cancer. An improved molecular understanding of cSCC may facilitate our ability to identify the best targets for precision therapy and improve the clinical outcomes in patients suffering from this recurrent and invasive malignancy.

Footnotes and Disclosures

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HUMAN SUBJECTS: Human subjects were included in this study. The study was approved by the University of Michigan Institutional Review Board and adhered to the tenets of the Declaration of Helsinki.[17] All participants provided written informed consent prior to enrollment in the study.

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Author Contributions:

Conception and design: H. Demirci, Elner, Chinnaiyan, F.Y. Demirci

Data collection: H. Demirci, Wu, Chinnaiyan, Robinson

Analysis and interpretation: H. Demirci, Vo, Wu, Elner, Robinson, F.Y. Demirci

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Abbreviations and Acronyms:
CNA = copy number alteration; **cSCC** = conjunctival squamous cell carcinoma; **FFPE** = formalin-fixed paraffin-embedded; **HPV** = human papillomavirus; **MSI** = microsatellite instability; **Mut/Mb** = mutations per megabase; **NGS** = next-generation sequencing; **OSSN** = ocular surface squamous neoplasia; **TMB** = tumor mutation burden; **UV** = ultraviolet; **WES** = whole-exome sequencing.

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References

1. Shields CL, Demirci H, Karatza E, Shields JA. Clinical survey of 1643 melanocytic and nonmelanocytic conjunctival tumors. *Ophthalmology*. 2004;111:1747–1754.
2. Gichuhi S, Sagoo MS, Weiss HA, Burton MJ. Epidemiology of ocular surface squamous neoplasia in Africa. *Trop Med Int Health*. 2013;18:1424–1443.
3. Lee GA, Hirst LW. Retrospective study of ocular surface squamous neoplasia. *Aust N Z J Ophthalmol*. 1997;25:269–276.
4. Gichuhi S, Macharia E, Kabiru J, et al. Risk factors for ocular surface squamous neoplasia in Kenya: a case-control study. *Trop Med Int Health*. 2016;21:1522–1530.
5. Yeoh CHY, Lee JJR, Lim BXH, et al. The management of ocular surface squamous neoplasia (OSSN). *Int J Mol Sci*. 2022;24:713.
6. Di Girolamo N. Association of human papilloma virus with pterygia and ocular-surface squamous neoplasia. *Eye (Lond)*. 2012;26:202–211.
7. Ramberg I, Møller-Hansen M, Toft PB, et al. Human papillomavirus infection plays a role in conjunctival squamous cell carcinoma: a systematic review and meta-analysis of observational studies. *Acta Ophthalmol*. 2021;99:478–488.
8. Galor A, Karp CL, Sant D, et al. Whole exome profiling of ocular surface squamous neoplasia. *Ophthalmology*. 2016;123:216–217.e1.
9. Ramos-Betancourt N, Field MG, Davila-Alquisiras JH, et al. Whole exome profiling and mutational analysis of ocular surface squamous neoplasia. *Ocul Surf*. 2020;18:627–632.
10. Lazo de la Vega L, Bick N, Hu K, et al. Invasive squamous cell carcinomas and precursor lesions on UV-exposed epithelia demonstrate concordant genomic complexity in driver genes. *Mod Pathol*. 2020;33:2280–2294.
11. Ramberg I, Vieira FG, Toft PB, et al. Genomic alterations in human papillomavirus-positive and -negative conjunctival

squamous cell carcinomas. *Invest Ophthalmol Vis Sci*. 2021;62:11.
12. Gleber-Netto FO, Nagarajan P, Sagiv O, et al. Histologic and genomic analysis of conjunctival SCC in african and American cohorts reveal UV light and HPV signatures and high tumor mutation burden. *Invest Ophthalmol Vis Sci*. 2024;65:24.
13. Robinson DR, Wu YM, Lonigro RJ, et al. Integrative clinical genomics of metastatic cancer. *Nature*. 2017;548:297−303.
14. Demirci H, Demirci FY, Ciftci S, et al. Integrative exome and transcriptome analysis of conjunctival melanoma and its potential application for personalized therapy. *JAMA Ophthalmol*. 2019;137:1444−1448.
15. Niu B, Ye K, Zhang Q, et al. MSIsensor: microsatellite instability detection using paired tumor-normal sequence data. *Bioinformatics*. 2014;30:1015−1016.
16. Roberts SA, Lawrence MS, Klimczak LJ, et al. An APOBEC cytidine deaminase mutagenesis pattern is widespread in human cancers. *Nat Genet*. 2013;45:970−976.
17. World Medical Association Declaration of Helsinki. Ethical principles for medical research involving human subjects. *JAMA*. 2013;310:2191−2194.
18. Lane DP. Cancer. p53, guardian of the genome. *Nature*. 1992;358:15−16.
19. Hedberg ML, Berry CT, Moshiri AS, et al. Molecular mechanisms of cutaneous squamous cell carcinoma. *Int J Mol Sci*. 2022;23:3478.
20. Pacifico A, Leone G. Role of p53 and CDKN2A inactivation in human squamous cell carcinomas. *J Biomed Biotechnol*. 2007;2007:43418.
21. Dotto GP. Crosstalk of Notch with p53 and p63 in cancer growth control. *Nat Rev Cancer*. 2009;9:587−595.
22. Ateenyi-Agaba C, Dai M, Le Calvez F, et al. TP53 mutations in squamous-cell carcinomas of the conjunctiva: evidence for UV-induced mutagenesis. *Mutagenesis*. 2004;19:399−401.
23. Tornesello ML, Waddell KM, Duraturo ML, et al. TP53 codon 72 polymorphism and risk of conjunctival squamous cell carcinoma in Uganda. *Cancer Detect Prev*. 2005;29:501−508.
24. Joanna R, Renata Z, Witold P, et al. The evaluation of human papillomavirus and p53 gene mutation in benign and malignant conjunctiva and eyelid lesions. *Folia Histochem Cytobiol*. 2010;48:530−533.
25. Gelbard MK, Munger K. Human papillomaviruses: knowns, mysteries, and unchartered territories. *J Med Virol*. 2023;95:e29191.
26. Litwin TR, Clarke MA, Dean M, Wentzensen N. Somatic host cell alterations in HPV carcinogenesis. *Viruses*. 2017;9:206.
27. Sharpless NE. INK4a/ARF: a multifunctional tumor suppressor locus. *Mutat Res*. 2005;576:22−38.
28. Porcheri C, Mitsiadis TA. Notch in Head and neck cancer. *Adv Exp Med Biol*. 2021;1287:81−103.
29. Reichrath J, Reichrath S. The impact of Notch signaling for carcinogenesis and progression of nonmelanoma skin cancer: lessons learned from cancer stem cells, tumor angiogenesis, and beyond. *Adv Exp Med Biol*. 2021;1287:123−154.
30. Chen ZG, Saba NF, Teng Y. The diverse functions of FAT1 in cancer progression: good, bad, or ugly? *J Exp Clin Cancer Res*. 2022;41:248.
31. Peng Z, Gong Y, Liang X. Role of FAT1 in health and disease. *Oncol Lett*. 2021;21:398.
32. Dauch C, Shim S, Cole MW, et al. KMT2D loss drives aggressive tumor phenotypes in cutaneous squamous cell carcinoma. *Am J Cancer Res*. 2022;12:1309−1322.
33. Kumar M, Molkentine D, Molkentine J, et al. Inhibition of histone acetyltransferase function radiosensitizes CREBBP/EP300 mutants via repression of homologous recombination, potentially targeting a gain of function. *Nat Commun*. 2021;12:6340.
34. Pickering CR, Zhou JH, Lee JJ, et al. Mutational landscape of aggressive cutaneous squamous cell carcinoma. *Clin Cancer Res*. 2014;20:6582−6592.
35. Yilmaz AS, Ozer HG, Gillespie JL, et al. Differential mutation frequencies in metastatic cutaneous squamous cell carcinomas versus primary tumors. *Cancer*. 2017;123:1184−1193.
36. de Guillebon E, Jimenez M, Mazzarella L, et al. Combining immunotherapy with an epidrug in squamous cell carcinomas of different locations: rationale and design of the PEVO basket trial. *ESMO Open*. 2021;6:100106.
37. Vanhaesebroeck B, Perry MWD, Brown JR, et al. PI3K inhibitors are finally coming of age. *Nat Rev Drug Discov*. 2021;20:741−769.
38. Scholz SL, Thomasen H, Reis H, et al. Frequent TERT promoter mutations in ocular surface squamous neoplasia. *Invest Ophthalmol Vis Sci*. 2015;56:5854−5861.
39. Lin SY, Liao SL, Hong JB, et al. TERT promoter mutations in periocular carcinomas: implications of ultraviolet light in pathogenesis. *Br J Ophthalmol*. 2016;100:274−277.
40. Starita N, Buonaguro L, Buonaguro FM, Tornesello ML. Telomerase promoter mutations in human immunodeficiency virus-related conjunctiva neoplasia. *J Transl Med*. 2018;16:77.
41. Jayaraj P, Sen S, Saxena K, et al. Immunohistochemical and mutational status of telomerase reverse transcriptase in conjunctival squamous cell carcinoma. *Indian J Ophthalmol*. 2022;70:971−975.
42. Gao J, Pickett HA. Targeting telomeres: advances in telomere maintenance mechanism-specific cancer therapies. *Nat Rev Cancer*. 2022;22:515−532.
43. Chang E, Demirci H, Demirci FY. Genetic aspects of conjunctival melanoma: a review. *Genes (Basel)*. 2023;14:1668.
44. Ellingsen EB, Bjørheim J, Gaudernack G. Therapeutic cancer vaccination against telomerase: clinical developments in melanoma. *Curr Opin Oncol*. 2023;35:100−106.
45. Liu M, Zhang Y, Jian Y, et al. The regulations of telomerase reverse transcriptase (TERT) in cancer. *Cell Death Dis*. 2024;15:90.
46. Tornesello ML, Cerasuolo A, Starita N, et al. Reactivation of telomerase reverse transcriptase expression in cancer: the role of TERT promoter mutations. *Front Cell Dev Biol*. 2023;11:1286683.
47. Huang T, Chen X, Zhang H, et al. Prognostic role of tumor mutational burden in cancer patients treated with immune checkpoint inhibitors: a systematic review and meta-analysis. *Front Oncol*. 2021;11:706652.
48. Demirci H, Elner VM, Demirci FY, et al. Immunotherapy for conjunctival squamous cell carcinoma with orbital extension. *Ophthalmology*. 2021;128:801−804.
49. Serbest Ceylanoglu K, Guneri Beser B, Singalavanija T, et al. Targeted therapy and immunotherapy for advanced malignant conjunctival tumors: systematic review. *Ophthalmic Plast Reconstr Surg*. 2024;40:18−29.
50. Fan J, Goldberg H, Torres-Cabala CA, et al. Complete response to immunotherapy using pembrolizumab in a patient with conjunctival squamous cell carcinoma with nodal metastasis. *Ophthalmic Plast Reconstr Surg*. 2024;40:e171−e174.
51. Asnaghi L, Alkatan H, Mahale A, et al. Identification of multiple DNA copy number alterations including frequent 8p11.22 amplification in conjunctival squamous cell carcinoma. *Invest Ophthalmol Vis Sci*. 2014;55:8604−8613.

52. Vizcaino MA, Tabbarah AZ, Asnaghi L, et al. ADAM3A copy number gains occur in a subset of conjunctival squamous cell carcinoma and its high grade precursors. *Hum Pathol*. 2019;94:92–97.
53. Mishra DK, Veena U, Kaliki S, et al. Differential expression of stem cell markers in ocular surface squamous neoplasia. *PLoS One*. 2016;11:e0161800.
54. Moses MA, George AL, Sakakibara N, et al. Molecular mechanisms of p63-mediated squamous cancer pathogenesis. *Int J Mol Sci*. 2019;20:3590.
55. Yi M, Tan Y, Wang L, et al. TP63 links chromatin remodeling and enhancer reprogramming to epidermal differentiation and squamous cell carcinoma development. *Cell Mol Life Sci*. 2020;77:4325–4346.